

PHASE 1 — SYSTEM IDENTIFICATION REPORT

Mobile Manipulation Robot · DSMfinal.py · PyBullet Simulation

1 SYSTEM DESCRIPTION

The system is a **ground mobile manipulation robot** simulated in PyBullet at 240 Hz. A Clearpath **Husky UGV** (4-wheel differential drive, 38 kg) carries a **KUKA iiwa 7-DOF arm** rigidly mounted at 0.25 m above the chassis. The robot executes a full **pick-and-place mission**: plan a collision-free route with A*, navigate to an automatically chosen dock beside the sample cube, grasp the block via a fixed constraint, transit to the goal, and deposit the block on the green pad.

Navigation uses an **A* grid planner** (0.35 m cells, 0.76 m total clearance) with greedy path simplification, segment densification, and a **pure-pursuit heading controller** (IIR-smoothed, rate-limited). A **24-candidate dock ring** selects the approach point that minimises path length while orienting the rover toward the goal direction.

2 ASSUMPTIONS

- A1. Flat, rigid ground plane (plane.urdf); no terrain variation.
- A2. Physics fidelity fixed at $\Delta t = 1/240$ s; real-time playback in GUI mode.
- A3. Arm base is kinematically locked to rover chassis (JOINT_FIXED).
- A4. Grasp is ideal: JOINT_FIXED constraint at the end-effector, $F_{\text{max}} = 500$ N.
- A5. Sample block mass = 0.16 kg; all obstacle bodies are static (mass = 0).
- A6. Arena bounds are [-7.2, 7.2] m in x and y; planner rejects out-of-bounds cells.
- A7. Obstacle geometry (positions, radii) is fully known at planning time.
- A8. Mission timeout = 200 s wall-clock time.

3 SYSTEM INPUTS AND OUTPUTS

Cat.	Signal / Parameter	Type	Value / Range	Description
IN	Obstacle geometry	Static config	4 boxes + 2 cylinders	Known at startup; used to build A* cost map
IN	Sample block pose	Sim state poll	getBasePositionAndOrientation	World-frame position + quaternion, read each step
IN	Husky base pose	Sim state poll	pos (x,y,z), yaw (rad)	Used by planner, heading controller, arm IK
IN	A* planned path	Computed	List of (x,y) world points	astar_path() with 0.76 m clearance, then simplified
IN	Dock candidate ring	Computed	24 pts at $r = 1.10$ m	choose_sample_dock() selects best approach point
IN	Mission phase	FSM state	NAV_TO_SAMPLE/PICK/NAV_TO_GOAL/PLACE/DONE	Each action executes each tick
IN	Control gains	Config	BASE_SPEED=6, TURN_GAIN=2, DRIVE_SMOOTH=0.5	and arm parameters
OUT	Wheel velocities	Actuator	omega in [-12, 12] rad/s x4	Velocity-control torque to joints 2,3,4,5
OUT	Arm joint angles	Actuator	7 joints, position control	KUKA iiwa targets via Cartesian IK + smoothstep
OUT	Grasp constraint	Virtual	JOINT_FIXED, $F_{\text{max}}=500$ N	Created at PICK phase; removed at PLACE
OUT	Sample block pose	Sim outcome	Rigid-body with constraint	Block follows end-effector while grasped
OUT	Mission completion	Terminal state	Phase.DONE then disconnect	Immediate exit once block deposited at goal

4 FREE BODY / SIGNAL-FLOW DIAGRAM

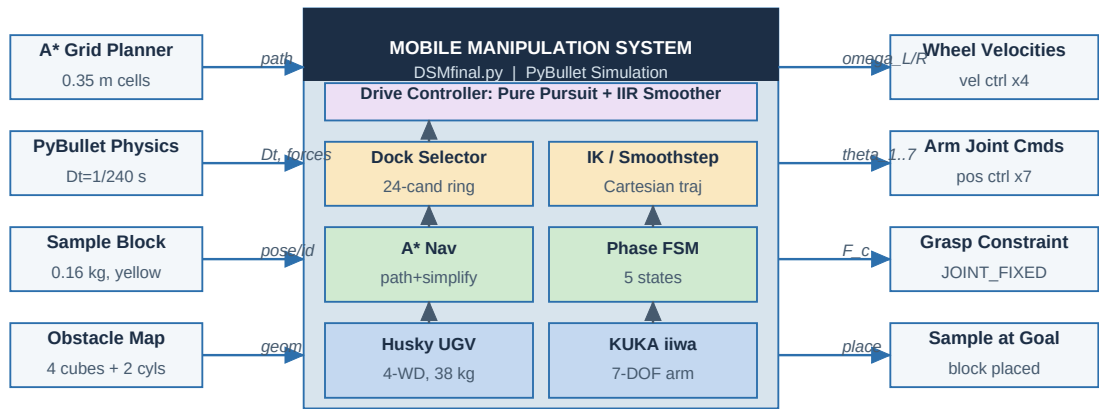


Figure 1 — Free Body / Signal-Flow Diagram

Navigation Parameter	Value	Arm / Sim Parameter	Value
Start position	(-6.0, -6.0, 0.1) m	Arm DOF	7 (KUKA iiwa)
Goal position	(+6.0, +6.0, 0.0) m	Arm mount height	0.25 m above chassis
Grid resolution	0.35 m / cell	IK iterations	180 max, eps=1e-4
Robot clearance	0.48 + 0.28 = 0.76 m	Pos gain / Vel gain	0.25 / 0.65
Dock ring radius	1.10 m from sample	Max joint velocity	1.2 rad/s
Dock candidates	24 (15° spacing)	Max joint force	160 N
Path reach threshold	0.65 m	Grasp constraint force	500 N
Cruise wheel speed	6.0 rad/s	Sim time-step	1/240 s, 2 sub-steps
Mission timeout	200 s	Solver iterations	200